DRAWING DESCRIPTIONS — PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture block diagram showing the neural processing system with the cognitive control layer, fallback controller, and the transition mechanism between connected and autonomous operation.

FIG. 2 is a state transition diagram showing the three operational states and the conditions that trigger transitions between them.

FIG. 3 is a graph showing the energy-aware modulation curve that governs autonomous self-regulation behavior across four energy regions.

FIG. 4 is a signal diagram showing the output waveform of the autonomous

input generator including circadian base signal and attention bursts.

FIG. 5 is a structural diagram of the resynchronization payload transmitted to the cognitive control layer upon reconnection.

FIG. 6 is a set of timeline diagrams showing three recovery scenarios: normal reconnection, crash recovery, and clean shutdown.

FIG. 7 is an end-to-end signal flow diagram showing the complete system during both connected and autonomous operation, illustrating how the fallback mechanism seamlessly replaces external cognitive modulation with internal energy-aware self-modulation without interrupting neural processing or energy harvesting.

DETAILED DESCRIPTION OF THE DRAWINGS

FIG. 1 — System Architecture with Fallback

This figure shows the complete system architecture including the fallback mechanism.

Top section — Cognitive Control Layer:

- Cloud shape labeled "External Cognitive Control Layer (e.g., Claude)"

- Two arrows exiting downward:

- "Modulation commands (-1.0 to +1.0)"

- "Heartbeat signal (implicit via tool calls)"

- Label: "Provides intelligent cognitive decisions"

- Dashed outline indicating "May become unavailable"

Middle section — Transition Controller:

- Rectangle labeled "Heartbeat Watchdog"

- Inside: "Monitors time since last cognitive input"

- Inside: "Timeout threshold: 30 seconds"

- Inside: "Check interval: 5 seconds"

- Two output paths:

- Left arrow (solid): "Heartbeat active -> Connected Mode"

- Right arrow (dashed): "Heartbeat timeout -> Autonomous Mode"

Left path — Connected Mode:

- Rectangle labeled "Cognitive-Driven Operation"

- Inside: "Input: Claude-controlled (0-1)"

- Inside: "Modulation: Claude-controlled (-1 to +1)"

- Arrow down to SNN

Right path — Autonomous Mode:

- Rectangle labeled "Autonomous Fallback Controller"

- Inside: "Input: Self-generated (circadian + bursts)"

- Inside: "Modulation: Energy-aware (-0.4 to +0.3)"

- Arrow down to SNN

- Additional outputs:

- Arrow to "Step Buffer" (rectangle)

- Arrow to "State Snapshots" (cylinder/disk shape)

Bottom section — Neural Processing:

- Large rectangle labeled "Spiking Neural Network + Energy Harvester"

- Inside: "Continuous operation regardless of control source"

- Label: "THE SYSTEM NEVER STOPS"

Key annotation:

- "Seamless transition: SNN continues processing without interruption"

- "The control source changes; the computation continues"

FIG. 2 — State Transition Diagram

This figure shows a finite state machine with three states and labeled transitions.

Three states (large circles or rounded rectangles):

State 1 (left, bold outline):

- Label: "CONNECTED"

- Sublabel: "Cognitive layer active"

- Inside: "Input from cognitive layer"

- Inside: "Modulation from cognitive layer"

State 2 (right):

- Label: "AUTONOMOUS"

- Sublabel: "Fallback active"

- Inside: "Self-generated input"

- Inside: "Energy-aware modulation"

- Inside: "State buffering active"

State 3 (bottom):

- Label: "RECOVERING"

- Sublabel: "Resynchronization"

- Inside: "Transmitting buffer"

- Inside: "Restoring cognitive control"

Transitions (labeled arrows between states):

CONNECTED -> AUTONOMOUS:

- Arrow labeled: "Heartbeat timeout (>30s no tool call)"

- Condition: "last_tool_call_time > 30s"

AUTONOMOUS -> RECOVERING:

- Arrow labeled: "Cognitive layer reconnects (resync called)"

- Condition: "resync() tool invoked"

RECOVERING -> CONNECTED:

- Arrow labeled: "Resync complete, buffer transmitted"

- Condition: "Cognitive layer acknowledges state"

AUTONOMOUS -> AUTONOMOUS (self-loop):

- Arrow labeled: "Each autonomous step (up to 10,000)"

- Condition: "autonomous_steps < hard_cap"

AUTONOMOUS -> CONNECTED (direct, dashed):

- Arrow labeled: "Any tool call received (heartbeat reset)"

- Condition: "Direct reconnection without explicit resync"

Additional annotations:

- Start indicator pointing to CONNECTED: "Initial state"

- Note on AUTONOMOUS: "Snapshot every 50 steps"

- Note on AUTONOMOUS self-loop: "Hard cap: 10,000 steps"

FIG. 3 — Energy-Aware Modulation Curve

This figure shows the step function mapping energy level to modulation value.

X-axis: "System Energy Level (mWh)" ranging from 0 to 100+

Y-axis: "Autonomous Modulation Value" ranging from -0.5 to +0.5

Step function with four plateaus:

Plateau 1 (0 to 15 mWh):

- Modulation = -0.4

- Region shaded with dense hatching

- Label: "CRITICAL — Maximum conservation"

- Behavioral note: "Inhibit neural activity to preserve energy"

Plateau 2 (15 to 30 mWh):

- Modulation = -0.1

- Region shaded with light hatching

- Label: "LOW — Cautious operation"

- Behavioral note: "Slightly reduce activity"

Plateau 3 (30 to 80 mWh):

- Modulation = 0.0

- Region unshaded (clear)

- Label: "NORMAL — Neutral operation"

- Behavioral note: "Standard autonomous processing"

Plateau 4 (80+ mWh):

- Modulation = +0.3

- Region shaded with light hatching

- Label: "SURPLUS — Opportunistic exploration"

- Behavioral note: "Increase activity to utilize excess energy"

Vertical dashed lines at energy thresholds (15, 30, 80 mWh) with labels.

Annotation at bottom:

- "The system autonomously adjusts its processing intensity based on

available energy — mimicking biological metabolic regulation"

FIG. 4 — Autonomous Input Generator Output

This figure shows a time-series waveform of the self-generated input signal.

X-axis: "Autonomous Step Number" ranging from 0 to 500 (representative)

Y-axis: "Input Value" ranging from 0.0 to 1.0

Main waveform components:

Component 1 — Circadian Base Signal:

- Smooth sinusoidal curve

- Equation annotation: "base = 0.5 + 0.3 x sin(2 x pi x step / period)"

- Period labeled: "period = 100 steps"

- Amplitude range: 0.2 to 0.8

Component 2 — Attention Bursts:

- Sharp upward spikes superimposed on the base signal

- Occurring randomly at approximately 10% of steps

- Burst magnitude: +0.2 above base signal

- Each burst labeled with a small arrow: "10% probability burst"

Component 3 — Composite Signal:

- The sum of base + bursts, shown as the final bold line

- Label: "Composite autonomous input"

Below the main graph, a smaller graph showing the corresponding energy level over the same time period:

- Shows energy fluctuating based on activity

- Annotations showing where modulation changes (at 15, 30, 80 mWh thresholds)

- Label: "Energy-aware modulation applied to composite input"

Annotation:

- "Circadian rhythm provides structured temporal variation"

- "Random bursts prevent the system from settling into trivial attractors"

- "Energy-aware modulation adjusts overall intensity"

FIG. 5 — Resynchronization Payload Structure

This figure shows the hierarchical data structure of the resync payload.

Top-level structure (large rectangle):

- Label: "Resynchronization Payload"

Three main sections within the payload:

Section 1 — Summary Statistics:

- Rectangle with fields:

- "steps_autonomous: [integer]" — Number of steps run autonomously

- "energy_delta: [float] mWh" — Net energy change during fallback

- "min_energy: [float] mWh" — Minimum energy observed

- "max_energy: [float] mWh" — Maximum energy observed

- "mean_energy: [float] mWh" — Average energy level

- "total_spikes: [integer]" — Sum of all spikes during fallback

- "mean_spikes_per_step: [float]" — Average spike rate

- Label: "ALWAYS included (compact summary)"

- Size annotation: "~200 bytes"

Section 2 — Energy Delta Detail:

- Rectangle with field:

- "energy_start: [float] mWh"

- "energy_end: [float] mWh"

- "energy_delta: [float] mWh"

- Arrow from energy_delta to annotation: "Positive = system gained energy during autonomy"

Section 3 — Full Step Buffer (Optional):

- Larger rectangle with repeating row structure:

- "Step 0: {input, modulation, spikes, energy, temp, pattern}"

- "Step 1: {input, modulation, spikes, energy, temp, pattern}"

- "..."

- "Step N: {input, modulation, spikes, energy, temp, pattern}"

- Label: "OPTIONAL — Full operational history"

- Size annotation: "~100 bytes x N steps"

- Condition: "Included when cognitive layer requests detailed history"

Two granularity levels annotated:

- Arrow to Section 1: "Level 1: Summary only (fast resync)"

- Arrow to Section 3: "Level 2: Full buffer (complete reconstruction)"

Decision flow at bottom:

- Diamond: "Cognitive layer requests full buffer?"

- Yes arrow: "Transmit Level 2 (summary + full buffer)"

- No arrow: "Transmit Level 1 (summary only)"

FIG. 6 — Recovery Timeline Diagrams

This figure shows three horizontal timeline diagrams stacked vertically, each illustrating a different recovery scenario.

Timeline A — Normal Reconnection:

Time -->

|-- Connected --|-- Timeout (30s) --|-- Autonomous --|-- Resync --|-- Connected -->
[Claude [Watchdog [Self-generated [Buffer [Cognitive

active] counting] input + energy transmitted] control

modulation] restored]

Labels on timeline:

- "Claude goes silent" (at transition to timeout)

- "Fallback engages" (at transition to autonomous)

- "resync() called" (at transition to resync)

- "Seamless resumption" (at return to connected)

- Buffer icon: "N steps buffered"

Timeline B — Crash Recovery:

Time -->

|-- Connected --|-- CRASH --|-- Server Restart --|-- Snapshot Load --|-- Connected -->
[Normal [MCP server [Process [Reads [State

operation] terminates] restarts] latest.json] restored]

Labels on timeline:

- "Server process dies" (at crash)

- "Process relaunched" (at restart)

- "initialize_consciousness loads snapshot" (at snapshot load)

- "State restored from last snapshot (max 50 steps lost)" (annotation)

- Snapshot icon: "latest.json (serialized every 50 steps)"

Timeline C — Clean Shutdown:

Time -->

|-- Connected --|-- Autonomous --|-- EOF Signal --|-- Shutdown -->

[Normal [Fallback [stdin closed] [Final snapshot

operation] operation] written to disk]

Labels on timeline:

- "Claude disconnects" (at transition to autonomous)

- "Process receives EOF" (at EOF signal)

- "finally block writes snapshot" (at shutdown)

- "State preserved for next session" (annotation)

Common annotations for all three timelines:

- Solid line segments: "System actively processing"

- Dashed line segments: "System in transition"

- Bold marker at each state change: "No data loss at any transition"

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous Operation

This figure shows the complete system signal flow in a split-view format,

with connected operation on top and autonomous operation on bottom, sharing the same core processing pipeline.

Layout — Two parallel horizontal flows sharing a common center:

TOP HALF — Connected Operation:

Label in top-left corner: "CONNECTED MODE"
[External Cognitive Layer] — cloud shape at top

|

v (dashed arrow: "Cognitive modulation commands")
[Sensors] -> [SNN] -> [Cognitive Modulation] -> [Motor Actuator] -> [Energy Harvester]

^ |

|_________ Energy feedback (bold dashed) ___________________|

^

|_________ Self-observation feedback (dotted) ___ from SNN output

Label on Cognitive Modulation block: "External AI sets reflection_coeff, modulation"

Label: "Heartbeat active (tool calls within 30s)"

DIVIDING LINE — Horizontal dashed line across full width

Label on line: "DISCONNECTION EVENT (heartbeat timeout > 30s)"

Arrow pointing down: "Seamless transition — no interruption to SNN or Harvester"

BOTTOM HALF — Autonomous Operation:

Label in bottom-left corner: "AUTONOMOUS MODE"

[Autonomous Input Generator] — rectangle replacing Sensors

Label inside: "Circadian base signal"

Label inside: "10% stochastic attention bursts"

|

v

[SNN] -> [Energy-Aware Self-Modulation] -> [Motor Actuator] -> [Energy Harvester]

^ |

|_________ Energy feedback (bold dashed) _________________________|

^

|_________ Self-observation feedback (dotted) ___ from SNN output

Label on Self-Modulation block: "Internal energy-aware regulation"

Four modulation regions shown as small annotation:

"<15 mWh: -0.4 | <30 mWh: -0.1 | 30-80 mWh: 0.0 | >80 mWh: +0.3"

Additional components in autonomous mode:

[State Buffer] — cylinder/database shape

Arrow from SNN: "Every step buffered"

Label: "Rolling buffer (max 10,000 entries)"

[Snapshot Writer] — disk shape

Arrow from State Buffer: "Every 50 steps"

Label: "latest.json (persistent storage)"

RIGHT SIDE — Reconnection:

Arrow from bottom half back to top half

Label: "resync() called"

[Resync Payload] — rectangle

Label inside: "steps_autonomous, energy_delta, summary"

Label inside: "Optional: full step-by-step buffer"

Arrow up to External Cognitive Layer: "Cognitive layer catches up"

Shared components (drawn once, spanning both halves):

The SNN, Motor Actuator, and Energy Harvester are the SAME blocks

in both halves — drawn in the center spanning the dividing line

Label: "Core pipeline never stops"

Key annotations:

"The fallback system changes the INPUT SOURCE and MODULATION SOURCE"

"The core neural processing, motor actuation, and energy harvesting"

"continue WITHOUT INTERRUPTION through the transition"

Legend box:

"Solid arrows: Forward signal path"

"Bold dashed: Energy feedback loop"

"Dotted: Self-observation feedback loop"

"Thin dashed: Cognitive/modulation control path"

"The transition between modes changes only the control paths,"

"never the core processing paths"

Sheet numbering: "7/7" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/7", "2/7", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures